

Place Value, Rounding and Estimating — Paper C

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to come out exactly, or to be estimated.
- The rules for powers of ten, rounding and significant figures are not printed on this paper. You are expected to know them.
- When you estimate, round every number to 1 significant figure and show what you rounded each one to — that is usually the first mark.
- Several questions ask whether an estimate is too big or too small. Track what each rounding did to the answer; if the two roundings pull in opposite directions, saying so is a correct answer.
- The mark for each question is shown in square brackets on the right.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – CHAINS AND MISSING MULTIPLIERS

1. Work out:

(a) $5 \times 0.1 \times 0.1$ [1]

(b) $3 \div 0.1 \div 0.1$ [1]

(c) $8 \times 0.1 \div 0.01$ [1]

(d) $0.6 \div 0.1 \times 0.01$ [1]

2. Fill in the missing number in each.

(a) $7.2 \times \underline{\hspace{1cm}} = 0.072$ [1]

(b) $0.9 \div \underline{\hspace{1cm}} = 90$ [1]

(c) $45 \times \underline{\hspace{1cm}} = 4.5$ [1]

(d) $0.03 \div \underline{\hspace{1cm}} = 3$ [1]

3. Round 0.0049682 to:

(a) 2 decimal places [1]

(b) 3 decimal places [1]

(c) 2 significant figures [1]

(d) 3 significant figures [1]

SECTION 2 – WHY THE TWO KINDS OF ROUNDING DIFFER

4. This question is about the number 0.04728.

- (a)** Round it to 2 decimal places and to 2 significant figures, then explain carefully [2]
why the two answers are different.

- (b)** Write down a number for which rounding to 2 decimal places and rounding to [2]
2 significant figures give the same answer, and explain why it does.

5. This question is about $8237 \div 38$.

- (a)** Estimate the answer by rounding each number to 1 significant figure. [1]

- (b)** The exact answer, to 1 decimal place, is 216.8. State whether your estimate is [1]
bigger or smaller than this.

- (c)** Explain why your estimate came out on that side, referring to what each [2]
rounding did to the answer.

SECTION 3 – WHAT A ROUNDED NUMBER REALLY TELLS YOU

6. A length is measured as 8.4 cm, correct to 1 decimal place.

- (a)** Write down the smallest value the true length could be. [1]

(b) Describe carefully the largest value the true length could be. [2]

(c) A second length is 12 cm, correct to the nearest whole number. Between what two values does this length lie? [2]

7. A student uses a calculator to work out 48.6×0.37 and writes down 179.82.

(a) Estimate 48.6×0.37 by rounding each number to 1 significant figure. [2]

(b) Use your estimate to show that 179.82 must be wrong, and suggest what the student probably typed in. [2]

(c) Write down the correct answer. [1]

End of paper. Check every answer has a unit or a form the question actually asked for.